

ALGORITHM
Spring 2026, Quiz 9, May 13
11:45-12:05 am

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1. Consider a variant of the matrix-chain multiplication problem in which the goal is to parenthesize the sequence of matrices so as to maximize, rather than minimize, the number of scalar multiplications.
 - (a) Does this problem exhibit optimal substructure?
 - (b) Find an optimal parenthesization of matrix-chain product whose sequence of dimensions is $\langle 10, 3, 12, 5, 50 \rangle$.
 - (c) Give an algorithm to solve this problem.